机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L6 Feedback Control Design via Pole Placement

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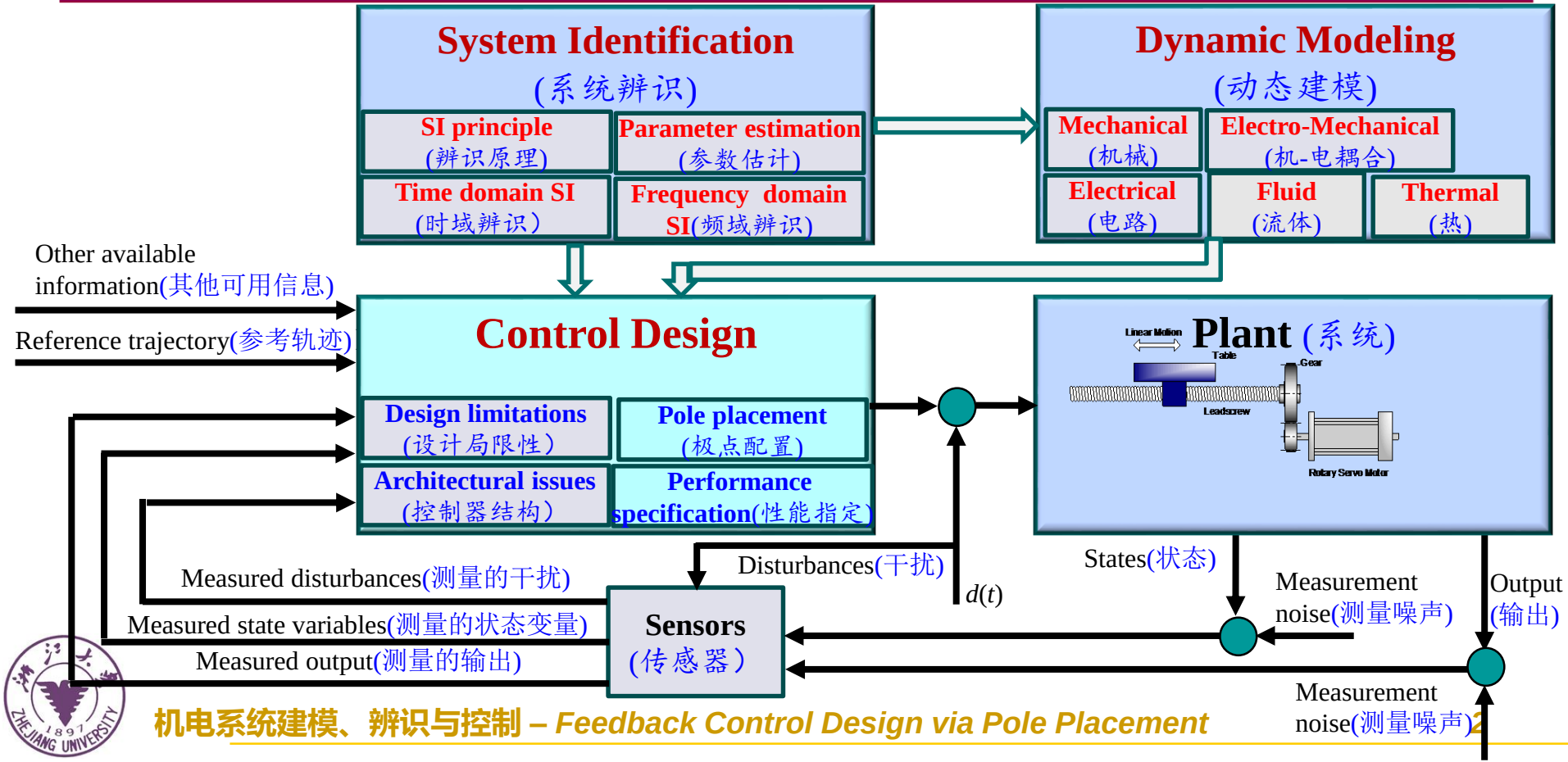
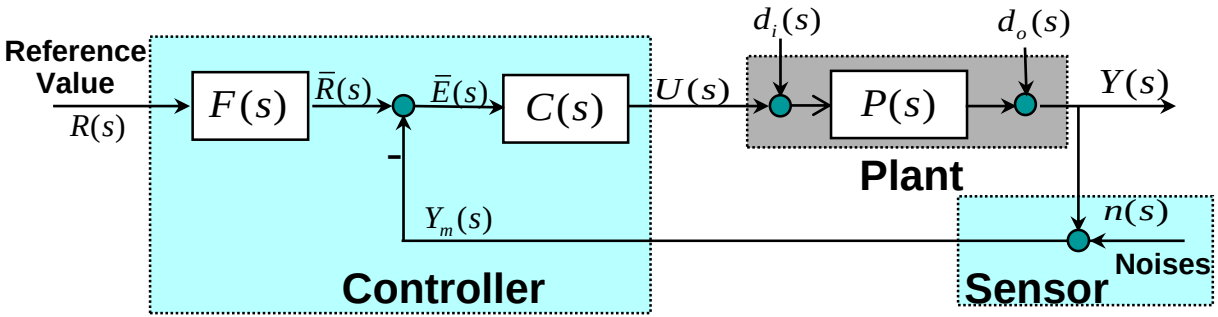
Main Contents of Course(课程内容)



Fundamental Problems in Control Design

- How to specify the desired performance that will be realized by closed-loop control ?
- Is there a uniform procedure for control design?
- Can any desired performances be realized via control design?
- Are there any limitation in control design?
- How to regard the difference of theoretical design and actual implementation?

Tracking Error of Two-DOF Controller



$L(s) = P(s)C(s)$, open-loop TF

- **CL Response in s-Domain**

$$Y(s) = \underbrace{T(s)F(s)}_{T_r(s)} R(s) + S_i(s)d_i(s) + S(s)d_o(s) - T(s)n(s)$$

$$S(s) = \frac{1}{1+L(s)} = \frac{D_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{N_p(s)N_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$S_i(s) = \frac{P(s)}{1+L(s)} = \frac{N_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

- **Tracking Error in s-Domain**

$$E(s) = R(s) - Y(s)$$

$$= \underbrace{\left(1 - T_r(s)\right)R(s)}_{E_r(s)} \quad \underbrace{-S_i(s)d_i(s)}_{E_{di}(s)} \quad \underbrace{-S(s)d_o(s)}_{E_{do}(s)} \quad \underbrace{+T(s)n(s)}_{E_n(s)}$$

error due to reference input | error due to input disturbance | error due to output disturbance | error due to measurement noise

Steady-State Error in Time Domain

- **Constant Steady-State Error**(定常稳态误差)

  By assuming that conditions of the Final Value Theorem (FVT) are satisfied, then, the tracking error $e(t)$ converges to the constant steady-state value e_{ss} given by (假设满足终值定理，那么跟踪误差收敛于恒定的稳态值)

$$e_{ss} = \lim_{s \to 0} sE(s)$$

- **Steady-State Error for Sinusoidal Inputs**(正弦输入的稳态误差)

$$E(s) = R(s) - Y(s) = \underbrace{\left(1 - T_r(s)\right)R(s)}_{\substack{E_r(s) \\ \text{error due to} \\ \text{reference input}}} \underbrace{-S_i(s)d_i(s)}_{\substack{E_{di}(s) \\ \text{error due to} \\ \text{input disturbance}}} \underbrace{-S(s)d_o(s)}_{\substack{E_{do}(s) \\ \text{error due to} \\ \text{output disturbance}}} \underbrace{+T(s)n(s)}_{\substack{E_n(s) \\ \text{error due to} \\ \text{measurement noise}}}$$

$$\boxed{\begin{aligned} r(t) &= A_r \sin(\omega_r t), \\ d_i(t) &= A_{di} \sin(\omega_d t), \\ d_o(t) &= A_{do} \sin(\omega_d t), \\ n(t) &= A_n \sin(\omega_n t) \end{aligned}}$$

$$e_{ss}(t) = \underbrace{\left|1 - T_r(j\omega_r)\right| A_r \sin\left(\omega_r t + \angle\left(1 - T_r(j\omega_r)\right)\right)}_{e_{rss}(t)} \underbrace{-\left|S_i(j\omega_d)\right| A_{di} \sin\left(\omega_d t + \angle S_i(j\omega_d)\right)}_{e_{diss}(t)}$$

$$\Rightarrow \underbrace{-\left|S(j\omega_d)\right| A_{do} \sin\left(\omega_d t + \angle S(j\omega_d)\right)}_{e_{doss}(t)} \underbrace{+\left|T(j\omega_n)\right| A_n \sin\left(\omega_n t + \angle T(j\omega_n)\right)}_{e_{nss}(t)}$$

System Type wrt Reference Input /Output Disturbance

For unity feedback systems, **system type with respect to reference input or output disturbance** is determined by **the number of integrators in its feedback path open-loop TF $L(s)$, i.e.,** (对于单位反馈系统，其相对于参考输入或输出扰动的系统类型由其反馈路径开环传递函数 $L(s)$中的积分器数量决定)

$$L(s) = \frac{1}{s^k} L_I(s),\ 0 \neq L_I(0) < \infty$$

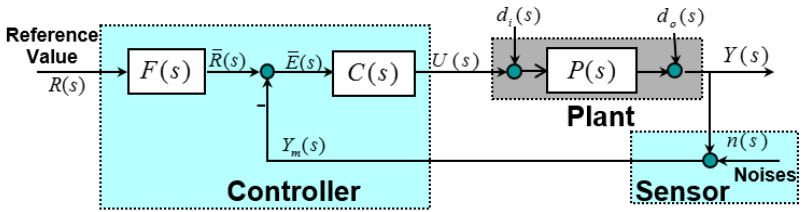
$$L(s) = ?$$

steady-state errors for unity output disturbance

	Step	Ramp	Parabola
Type 0	$-1/(1+K_p)$	∞	∞
Type 1	0	$-1/K_{dv}$	∞
Type 2	0	0	$-1/K_{da}$

steady-state errors for unity reference input

	Step	Ramp	Parabola
Type 0	$1/(1+K_p)$	∞	∞
Type 1	0	$1/K_v$	∞
Type 2	0	0	$1/K_a$



$$K_p = L(0)$$ position error constant

$$K_{dv} = \lim_{s \to 0} sL(s) = L_I(0)$$ velocity error constant

$$K_{da} = \lim_{s \to 0} s^2 L(s) = L_I(0)$$ acceleration error constant

System Type wrt Input Disturbance

For unity feedback systems, system type with respect to input disturbance is determined by **the number of integrators in its feedback path open-loop TF** $C(s)$, **i.e.,**(对于单位反馈系统，其输入干扰对应的系统类型由反馈路径开环传递函数C(s)中的积分器数量决定)

$$C(s)=\frac{1}{s^k}C_I(s),\ \ 0\neq C_I(0)<\infty$$

the steady-state errors for unity polynomial input disturbances are summarized by the following table

	Step	Ramp	Parabola
Type 0	$-1/K_{dp}$	∞	∞
Type 1	0	$-1/K_{dv}$	∞
Type 2	0	0	$-1/K_{da}$

$$K_{dp}=C(0)+\frac{1}{P(0)}$$

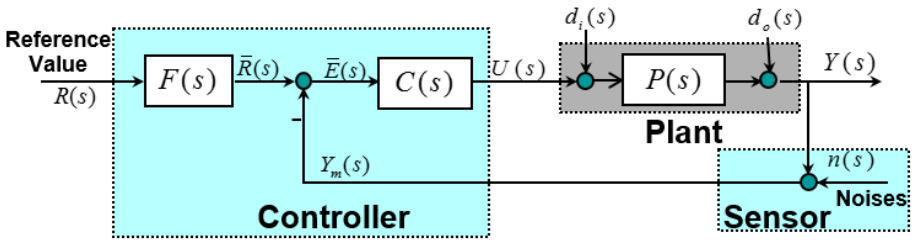
input disturbance position constant

$$K_{dv}=\lim_{s\to 0}sC(s)=C_I(0)$$

input disturbance velocity constant

$$K_{da}=\lim_{s\to 0}s^2C(s)=C_I(0)$$

input disturbance acceleration constant



Performance Specifications

$$E(s) = \left(1 - T_r(s)\right)R(s) - S_i(s)d_i(s) - S(s)d_o(s) + T(s)n(s)$$

The objective of control system design is to choose the controller TF $C(s)$ and $F(s)$ such that certain performance (design) specifications are met. We want: (控制系统设计的目标是选择合适的控制器C(s)和F(s)，以便系统能满足一定的性能指定, 我们希望：)

(i) The output $y(t)$ to follow the reference input $r(t)$(输出$y(t)$跟踪参考输入$r(t)$)

Ideally, want

$$T_r(s) \equiv 1$$

$$T_r(s) = T(s)F(s)$$

In reality, try to achieve

$$T_r(j\omega) \approx 1, \quad \omega \in \underbrace{\left[\omega_{low}, \quad \omega_{high}\right]}_{\text{Frequency Range of Ref. Input}}$$

Performance Specifications

$$E(s)=\left(1-T_r(s)\right)R(s)-S_i(s)d_i(s)-S(s)d_o(s)+T(s)n(s)$$

(ii) The disturbances to have very little effect on the output $y(t)$ within the frequency range where disturbances are most likely to occur（干扰在其发生的频段上对系统输出具有较小的影响）

Ideally, want $S(s)\equiv 0$ $\quad S_i(s)=P(s)S(s)$

In reality, try to achieve $S(j\omega)\approx 0,\quad \omega\in\underbrace{\left[\omega_{d_low},\quad \omega_{d_high}\right]}_{\text{Frequency Range of Disturbances}}$

(iii) The noise to have very little effect on the output $y(t)$ within the frequency range where noises are most likely to occur（噪声在其发生的频段上对系统输出具有较小的影响）

Ideally, want $T(s)\equiv 0$

In reality, try to achieve $T(j\omega)\approx 0,\quad \omega\in\underbrace{\left[\omega_{n_low},\quad \infty\right]}_{\text{Frequency Range of Noises}}$

Steady-State Performance Specifications

- **When Dealing with Polynomial Type Inputs in Time-Domain**
  - ➢Requirements for steady-state error

$$\left|\frac{r(t)-y_{SS}(t)}{r(t)}\right| \le X\%$$

Specific requirements on the system types and the corresponding error constants

Reference input	Input function	Requirement for system type w.r.t. reference input	Steady-state error	Requirement for corresponding error constants
Constant reference	$r(t)=r_{ss}$	Type 0	$\left\|\frac{1}{1+K_P}\right\|<X\%$	$K_P>\frac{1}{X\%}-1$
Ramp reference	$r(t)=t$	Type 1	$\left\|\frac{1}{K_V}\right\|<X\%$	$K_V>\frac{1}{X\%}$

Steady-State Performance Specifications

- **When Dealing with Sinusoidal Type Inputs in Frequency-Domain**
  - The steady-state error $e_{SS}(t)$ to track a reference input should be within $X\%$ and the steady-state error due to the input disturbance/output disturbance/noise should be attenuated by a factor γ (跟踪参考输入的稳态误差$e_{SS}(t)$应控制在X%以内，而由输入扰动/输出扰动/噪声引起的稳态误差应衰减至γ倍)

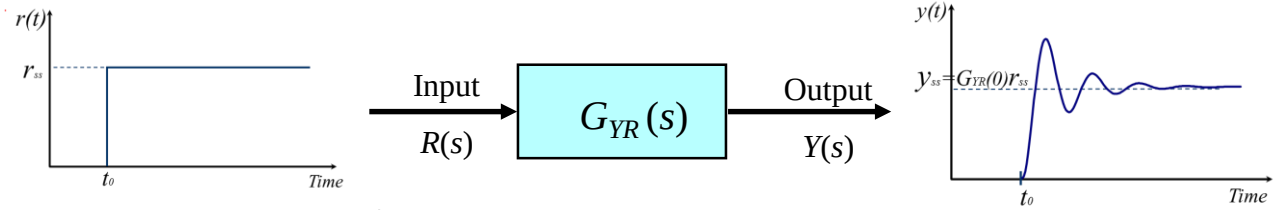
$$E(s)=\left(1-T_r(s)\right)R(s)-S_i(s)d_i(s)-S(s)d_o(s)+T(s)n(s)$$

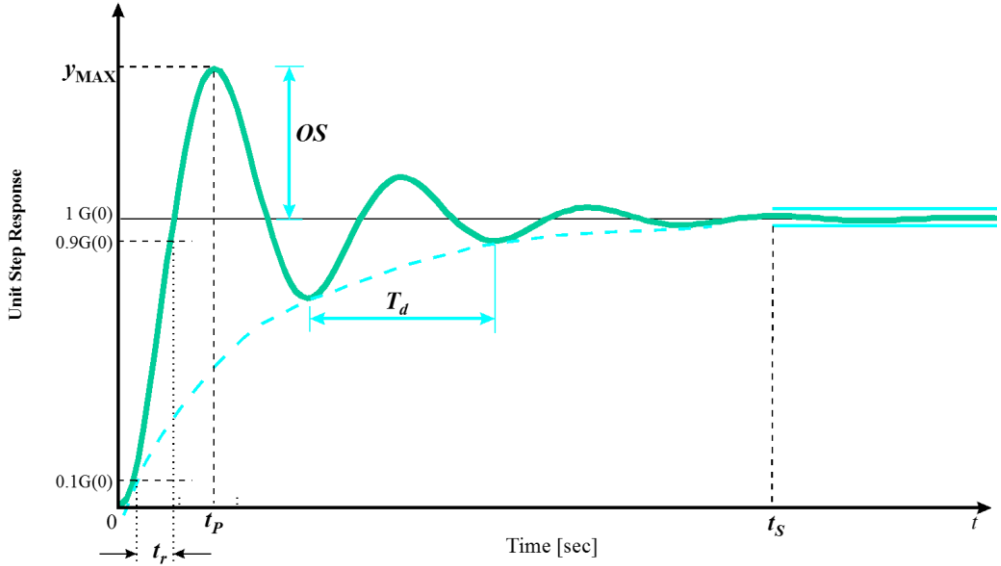
Specific requirements on the CL frequency responses

Reference input /disturbance/noise	function	Steady-state error	Requirement for sensitivity function
Reference input	$r(t)=A_r\sin(\omega_r t)$	$e_{rss}(t)=\lvert 1-T_r(j\omega_r)\rvert A_r\sin\left(\omega_r t+\angle\left(1-T_r(j\omega_r)\right)\right)$	$\lvert 1-T_r(j\omega_r)\rvert<X\%$
Input disturbance	$d_i(t)=A_{di}\sin(\omega_d t)$	$e_{diss}(t)=-\lvert S_i(j\omega_d)\rvert A_{di}\sin\left(\omega_d t+\angle S_i(j\omega_d)\right)$	$\lvert S_i(j\omega_d)\rvert<\frac{1}{\gamma}$
Output disturbance	$d_o(t)=A_{do}\sin(\omega_d t)$	$e_{doss}(t)=-\lvert S(j\omega_d)\rvert A_{do}\sin\left(\omega_d t+\angle S(j\omega_d)\right)$	$\lvert S(j\omega_d)\rvert<\frac{1}{\gamma}$
noise	$n(t)=A_n\sin(\omega_n t)$	$e_{nss}(t)=-\lvert T(j\omega_n)\rvert A_n\sin\left(\omega_n t+\angle T(j\omega_n)\right)$	$\lvert T(j\omega_n)\rvert<\frac{1}{\gamma}$

Transient Performance Specifications

- **Traditionally Specified Based on Unit Step Response**





Performance Specification → CL Pole Locations

- **Transient Performance Specifications and CLTF Pole Locations**

Transient performance specifications can be interpreted as constraints on the locations of the poles of the closed-loop transfer function. (瞬态性能指标可视为对闭环传递函数极点位置的约束条件)

$$p_{1,2} = -\sigma \pm j\omega_d = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

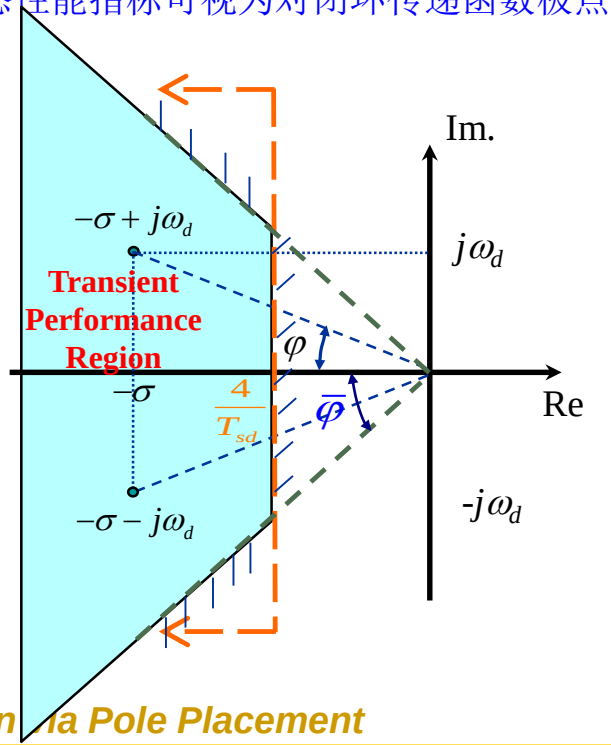
(1)Settling Time (2 %) ≤ T_{sd}

$$t_s(\pm 2\%) = \frac{4}{\xi\omega_n} = \frac{4}{\sigma} \le T_{sd} \Rightarrow \sigma \ge \frac{4}{T_{sd}}$$

(2) %OS ≤ X %

$$\%OS = 100\% \times e^{-\pi\cdot\frac{\sigma}{\omega_d}} \le X\%$$

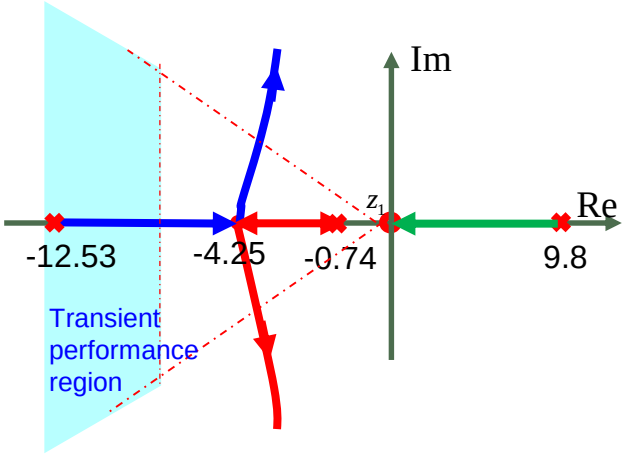
$$\Rightarrow \tan(\varphi) = \frac{\sqrt{1-\zeta^2}}{\zeta} \le -\frac{\pi}{\ln(X\%)} \Leftrightarrow \varphi \le \underbrace{\arctan\left(-\frac{\pi}{\ln(X\%)}\right)}_{\bar{\varphi}}$$



Examples of Unavailable Performance

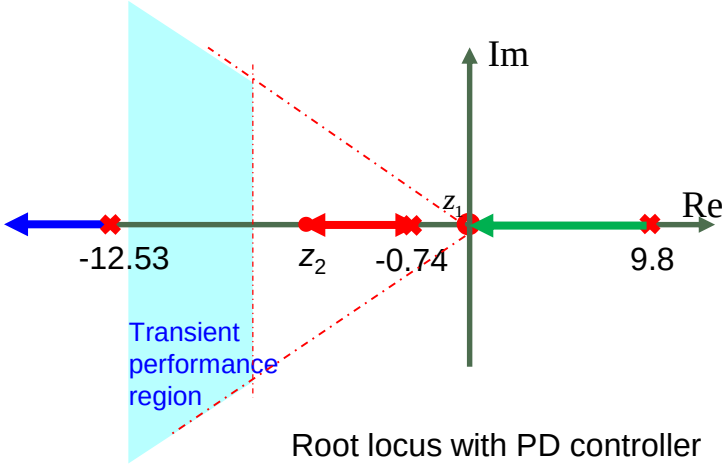
Plant model $P_\alpha(s) = \dfrac{9.962s}{(s+12.53)(s+0.7393)(s-9.8)}$

Q: Can a P controller stabilize the system?



Root locus with P controller

Q: How about a PD controller?

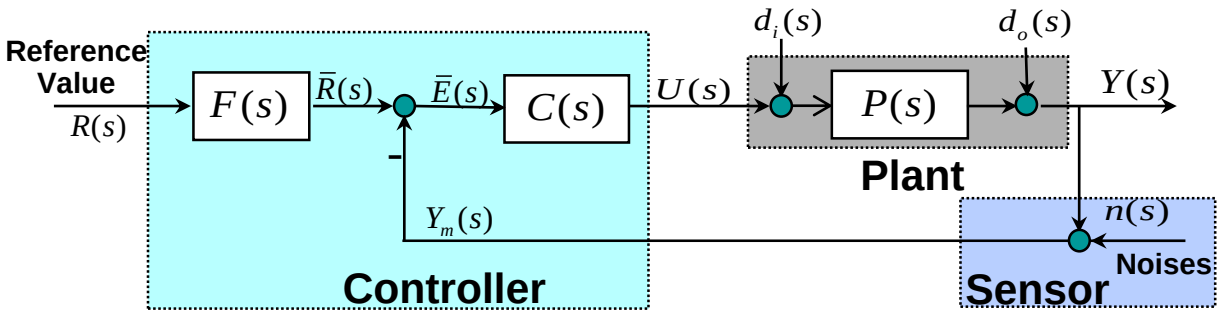


Root locus with PD controller

Please note the open-loop poles and zeros for placing the desired poles !!!

Feedback Control Design via Pole Placement

- **Motivation**
  - A systematic procedure to synthesize a controller for SISO LTI systems to place poles of nominal closed-loop system at certain desired locations（一种用于单输入单输出线性时不变系统控制器的系统化设计方法，可将名义闭环系统的极点置于特定的期望位置）



Real system $P(s)$

Can we obtain real system model? What model we used for control design?

Nominal system

$$P_n(s)=\frac{N_{pn}(s)}{D_{pn}(s)}=\frac{b_{pm}s^m+b_{p(m-1)}s^{m-1}+\cdots+b_{p1}s+b_{p0}}{s^n+a_{p(n-1)}s^{n-1}+\cdots+a_{p1}s+a_{p0}} \qquad m<n$$

Feedback controller

$$C(s)=\frac{N_c(s)}{D_c(s)}=\frac{b_{cm_c}s^{m_c}+b_{c(m_c-1)}s^{m_c-1}+\cdots+b_{c1}s+b_{c0}}{s^{n_c}+a_{c(n_c-1)}s^{n_c-1}+\cdots+a_{c1}s+a_{c0}} \qquad m_c<n_c$$

Pole Placement

- **Nominal Closed Loop Characteristic Equation**（名义闭环特征方程）

$$D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)=0$$

where $N_c(s)=b_{cm_c}s^{m_c}+b_{c(m_c-1)}s^{m_c-1}+\cdots+b_{c1}s+b_{c0}$

m_c+1 controller coefficients for N_c: $b_{cmc},\ldots,b_{c0}$

$$D_c(s)=s^{n_c}+a_{c(n_c-1)}s^{n_c-1}+\cdots+a_{c1}s+a_{c0}$$

n_c controller parameters to choose for D_c: $a_{c(nc-1)},\ldots,a_{c0}$

- **Nominal Closed Loop Characteristic Polynomial**（名义闭环特征多项式）

$$A_C(s)=D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)=s^{n_L}+a_{Cn_L-1}s^{n_L-1}+\cdots+a_{C1}s+a_{C0}$$

Order $m+m_c<n+n_c=n_L$

- **Desired Closed Loop Characteristic Polynomial**（期望闭环特征方程）

$$A_{Cd}(s)=(s-p_{1d})(s-p_{2d})\cdots(s-p_{n_Ld})=s^{n_L}+a_{C(n_L-1)d}s^{n_L-1}+\cdots+a_{C1d}s+a_{C0d}$$

Pole Placement

- **General Formulation of Pole Placement**(极点配置的一般公式)
  - Given $P_n(s)$ and any desired CL pole locations, i.e., known n_L and $A_{Cd}(s)$ but with coefficients being arbitrarily specified, does there exist a proper $C(s)$ that makes $A_C(s)$ being the desired $A_{Cd}(s)$?(给定Pn(s)和任何期望的闭环极点位置，即已知系数是任意指定的n_L和$A_{cd}(s)$，是否存在一个适当的C(s)使A_C成为所需的$A_{cd}(s)$？)

$$D_{pn}(s)D_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s)$$

SISO Polynomial Approach

Case 1: $n_L = 2n - 1 + k$

→ The controller order is $n_c = n-1+k, k>0$

Then, we have n_c controller parameters to choose for $D_c(s)$ and m_c+1 parameters for $N_c(s)$

→ A total of $n_c + m_c + 1 = n - 1 + k + m_c + 1$ coefficients free to choose

$$n_c + m_c + 1 \le 2n_c+1 = 2(n - 1 + k)+1 = 2n - 1 + 2k = n_L + k, \text{ (since } m_c \le n_c)$$

→ (i) The number of free controller parameters, $2n - 1+2k=n_L+k$, to match

→ (ii) But to match $A_c(s)$ with the desired $A_{cd}(s)$, we only have to make sure $a_{c(nL-1)},\ldots,$ a_{c0} equal to $a_{c(nK-1)},\ldots, a_{c0}$, which represent n_L equations to solve

→ As long as $k \ge 0$, infinite solution, when $k=0$, the solution is unique

Case 2: $n_L < 2n - 1$

→ Can not arbitrarily place the desired poles

PID Controller via Pole Placement

- **Proper PID Controller Structure**

$$C(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{\tau_D s + 1}$$

$$= K_P\left(1 + \frac{1}{T_r s} + \frac{T_d\, s}{\tau_D s + 1}\right)$$

$$= \frac{\left(K_D + K_P\tau_D\right)s^2 + \left(K_P + K_I\tau_D\right)s + K_I}{\tau_D s^2 + s}$$

$$T_r = \frac{K_P}{K_I}, \quad \text{Reset time}$$

$$T_d = \frac{K_D}{K_P}, \quad \text{Derivative time}$$

- **Equivalent Controller Form**

$$C(s) = \frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s^2 + a_{c1}s}$$

$$b_{c2} = \frac{K_D + K_p\tau_D}{\tau_D}, \quad b_{c0} = \frac{K_I}{\tau_D}$$

$$b_{c1} = \frac{K_p + K_I\tau_D}{\tau_D}, \quad a_{c1} = \frac{1}{\tau_D}$$

or

$$\tau_D = \frac{1}{a_{c1}}, \quad K_p = \left(b_{c1} - K_I\right)\tau_D$$

$$K_I = b_{c0}\tau_D, \quad K_D = \left(b_{c2} - K_p\right)\tau_D$$

PID Controller via Pole Placement

- **Determine PID Gains via Pole Placement**

$$D_{pn}(s)\left(s^2+a_{c1}s\right)+N_{pn}(s)\left(b_{c2}s^2+b_{c1}s+b_{c0}\right)=A_{Cd}(s)$$

Case 1: τ_D is pre-determined based on noise attenuation consideration

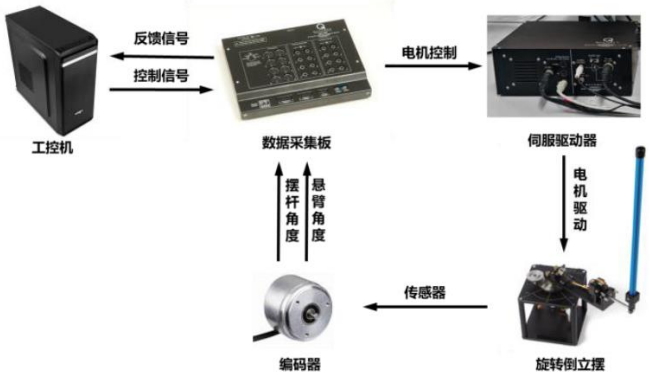
Only the Controller parameters b_{c2},b_{c1},b_{c0},or k_p,k_D and k_I are free to chosen. Thus, can only place three CL poles at the desired locations. So for plant with n>1,cannot arbitrarily place all CL poles. (只有控制器参数b_{c2}、b_{c1}、b_{c0}或k_p、k_D和k_I可以自由选择。故只能在期望位置放置三个闭环极点。因此，对于n>1的系统，不能任意放置所有闭环极点。)

Case 2: τ_D is automatically determined from pole placement design

Four controller parameters to choose, can place four CL poles at wish. So for plant with n≤2, we can arbitrarily place all CL poles.（可选择四个控制器参数，可以任意放置四个闭环极点。因此，对于n≤2的系统，我们可以任意配置所有闭环极点）

Balance Control of Inverted Pendulum

- 倒立摆系统组成
  - 倒立摆
  - 伺服电机
  - 齿轮减速器
  - 电流驱动器
  - **CRIO控制器**
  - 编码器

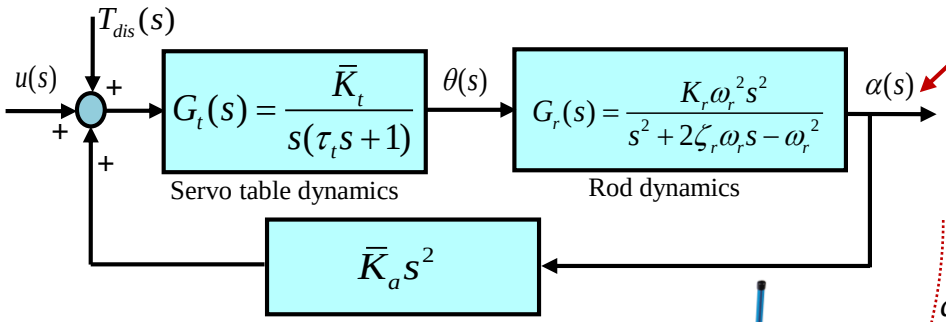


旋转倒立摆工作原理示意图

Physical parameters of nominal model

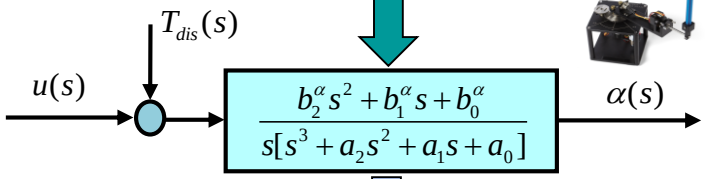
Symbol	Description	Value	Unit
m_p	Mass of pendulum	0.127	kg
L_c	Half of the total length of pendulum	0.1685	m
L_p	Total length of pendulum	0.337	m
J_p	Pendulum moment of inertia about center of mass	0.0012	kg.m^2
B_p	viscous damping coefficient of the pendulum	0.0024	N.m.s/rad
L_t	Rotary arm length from pivot to tip	0.216	m
J_{eq}	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2
B_{eq}	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad
R_m	Motor armature resistance	2.6	Ω
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)
K_g	Total gear ratio	14	
η_m	Motor efficiency	0.69	
η_g	Gearbox efficiency	0.9	
V_m	Applied voltage to the motor	-10~10	V
A_c	Coefficient of Coulomb friction torque	0.4	V

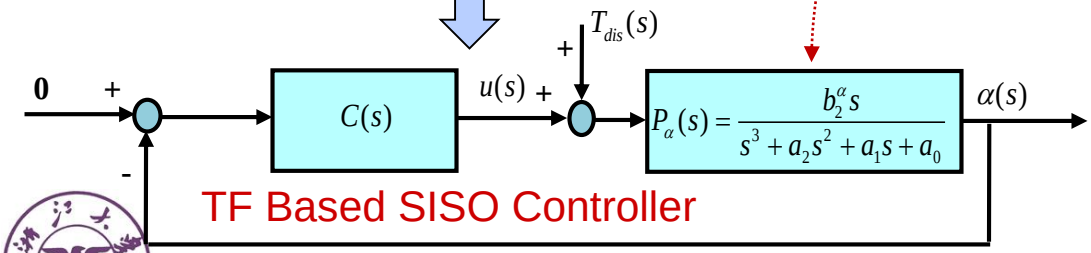
Output Stabilization of Inverted Pendulum



Let single output be the rod angle

$$\alpha(s) = [u(s) + T_{dis}(s) + \bar{K}_a s^2 \alpha(s)] \cdot G_t \cdot G_r$$





TF Based SISO Controller

$$P_\alpha(s) = \frac{\alpha(s)}{U(s)} = \frac{b_2^\alpha s^2 + b_1^\alpha s + b_0^\alpha}{s(s^3 + a_2 s^2 + a_1 s + a_0)}$$

$$a_0 = -\frac{\omega_t \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}, a_1 = \frac{2\zeta_r \omega_t \omega_r - \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}, a_2 = \frac{\omega_t + 2\zeta_r \omega_r}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2},$$

$$b_2^\alpha = \frac{\bar{K}_t K_r \omega_t \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}, b_0^\alpha = b_1^\alpha = 0, \omega_t = \frac{1}{\tau_t}$$

$$K_t = 4.9757, \omega_t = 0.7453, K_a = 0.18, A_c = 0.9,$$

$$\omega_r = 6.6059, \zeta_r = 0.0378, K_r = 0.022,$$

$$\Rightarrow s(s^3 + a_2 s^2 + a_1 s + a_0) = s(s^3 + 3.476s^2 - 120.8s - 90.83)$$

$$= s(s + 12.5371)(s + 0.7393)(s - 9.8)$$

$$b_2^\alpha = 9.962$$

$$\Rightarrow P_\alpha(s) = \frac{9.962s}{(s + 12.53)(s + 0.7393)(s - 9.8)}$$

Stabilization via Rod Angle Measurement Only

- **Performance Specifications**

a) No steady-state (SS) error for constant input disturbances

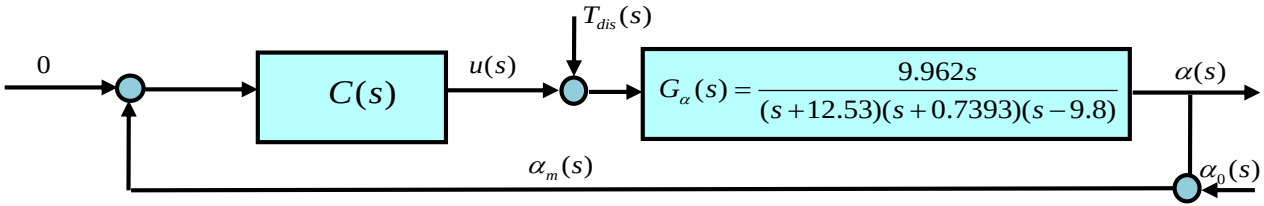
$\Rightarrow$ $L(s)$ need include an integrator

b) As fast as possible good transient response:

(b.1)Settling Time (2 %) $\leq$ 0.5s $\Rightarrow$ $\sigma = \omega_n \zeta \geq \dfrac{4}{T_{sd}} = 8$

(b.2) %OS $\leq$ 20 % $\Rightarrow$ $\dfrac{\sqrt{1-\zeta^2}}{\zeta} \leq -\dfrac{\pi}{\ln(X\%)} \Leftrightarrow \zeta \geq 0.4559$

Stabilization via Rod Angle Measurement Only



- **Controller Structure via Pole-Placement Design**
  - ✓ Spec. (a) is satisfied for placing two integrators in the denominator of controller (why?) (控制器分母配置两个积分项以满足条件(a))
  - ✓ Choosing n_c=3 for arbitrary pole-placement to meet Spec.(b) (why?)(选择n_c=3从而实现任意极点配置以满足条件(b)

$$P_\alpha(s)=\frac{9.962s}{(s+12.53)(s+0.7393)(s-9.8)}$$

$$C(s)=\frac{b_{c3}s^3+b_{c2}s^2+b_{c1}s+b_{c0}}{s^2(s+a_{c0})}$$

we can arbitrarily place all five CL poles: $p_{1,2d}$, p_{3d},p_{4d} p_{5d}

$$A_{CL}(s)=D_p(s)D_c(s)+N_p(s)N_c(s)$$

$$\Leftrightarrow A_{CLd}(s)=A_{CLd}(s)=\prod_{i=1}^{5}(s-p_{id})$$

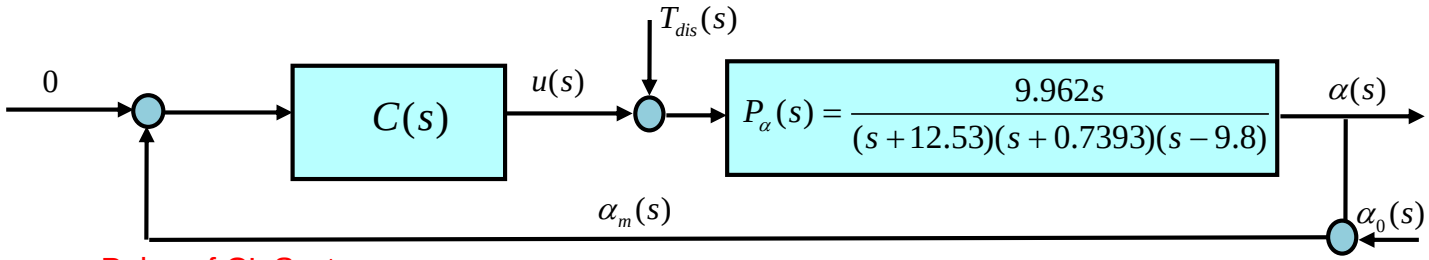
Dominate poles

$$p_{1,2d}=-\zeta\omega_n\pm j\omega_n\sqrt{1-\zeta^2}$$

$$\begin{cases}\omega_n\ge\dfrac{4}{\zeta T_{sd}}\\ \zeta\ge\bar{\zeta}==\left[\dfrac{\sqrt{1-\bar{\zeta}^2}}{\bar{\zeta}}=-\dfrac{\pi}{\ln(OS\%)}\right]\end{cases}$$

Stabilization via Rod Angle Measurement Only

- **Experimental Implementation**
  - Rod angle measurement is through an incremental encoder, leading to potential non-zero angle initialization or a constant measurement noise（摆杆角度测量通过增量编码器实现，可能导致初始角度非零或测量噪声恒定）
  - Coulomb friction effect is very significant as the target table speed is zero（当伺服转台速度为零时，库仑摩擦效应极为显著）



Desired Performance

ω_{nd}=13.6
ζ_d=0.59

Poles of CL System

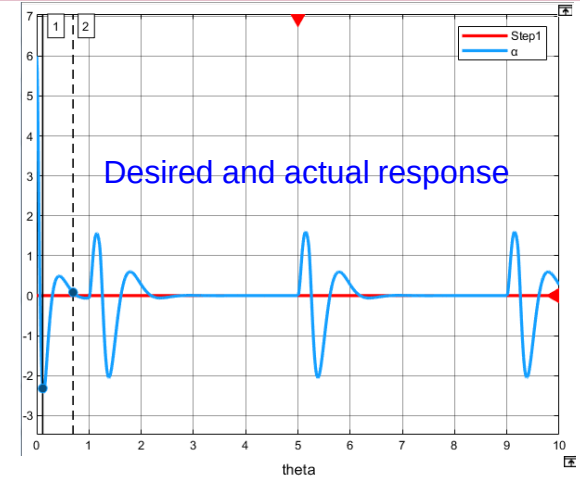
0.0000 + 0.0000i
-40.1792 + 0.1026i
-40.1792 - 0.1026i
-40.0016 + 0.0000i
-8.0240 +10.9807i
-8.0240 -10.9807i

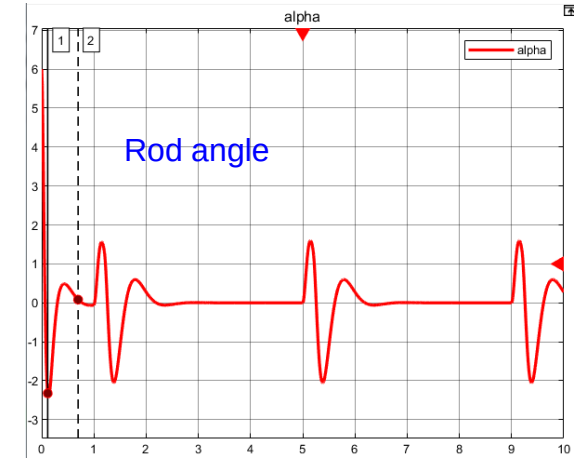
$$\Rightarrow C(s) = \frac{663s^3 + 1.812\times10^4 s^2 + 1.949\times10^5 s + 1.199\times10^6}{s^2(s+132.9)}$$

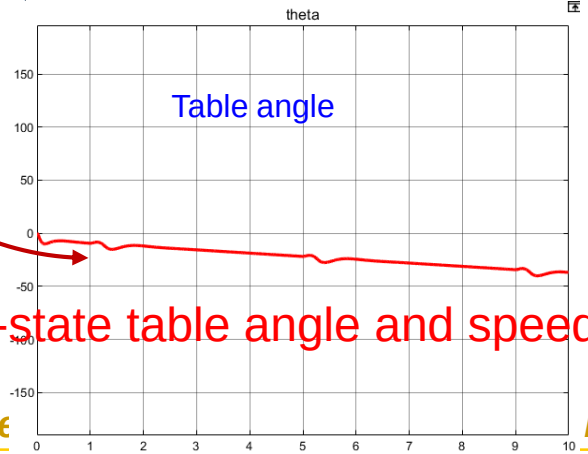
$$= \frac{659.05(s+16.03)(s+5.6512+j8.995)(s+5.6512-j8.995)}{s^2(s+132.9)}$$

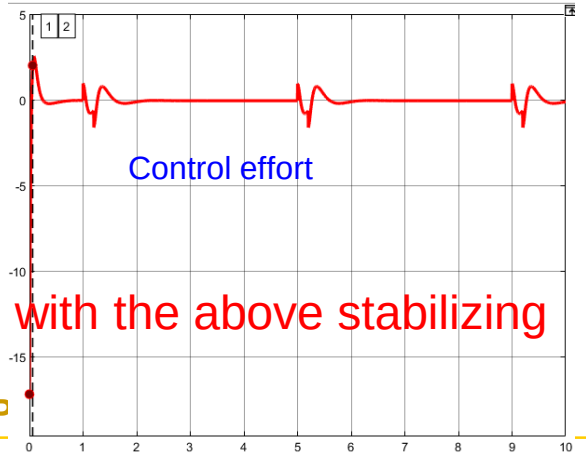
Stabilization via Rod Angle Measurement Only

- Simulation Results
  - ✓ with nonlinear models (A_c=0.9)
  - ✓ under pulse wave like input disturbance
  - ✓ initial rod table of 6deg
- Experimental Implementation
  - ✓ Potential instability due to table run-away; (why?)









Q: What are the steady-state table angle and speed with the above stabilizing controller?